



Robust and Fast Voltage Regulation Strategy of the Single-Ended Primary-Inductor Converter Using Modified Discrete-Time Sliding Mode Control

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Abstract: Efficient regulation of the direct-current (DC) voltage plays a vital role in increasing the stability and reliability of the renewable energy systems when they are operating under variable wind speed and solar irradiation. The single-ended primary-inductor converter (SEPIC) can operate efficiently in buck-boost mode without inverting the output voltage if a robust control technique is used to mitigate variation in the input voltage. This paper proposed a modified discrete-time sliding mode controller (MDTSMC) that can externally generate a reference switching variable trajectory generator to ensure fast and accurate voltage regulation for the SEPIC converter while minimizing the effects of disturbances and reducing the quasi-sliding-mode bandwidth. Initially, the effectiveness of the proposed approach is evaluated in a MATLAB/Simulink environment across four distinct test scenarios, demonstrating its capability to maintain voltage regulation under the influence of disturbances, unmodelled dynamics, and system parameter variations. To further validate the practical feasibility of the proposed strategy, hardware-in-the-loop (HIL) simulations are conducted by using the OPAL-RT platform under multiple operating cases. The HIL results confirm that the MDTSMC provides excellent dynamic response and resilience against load and input fluctuations, highlighting its suitability for real-time digital control implementation in SEPIC converters.

Keywords: Modified discrete-time sliding mode controller; Reference switching variable generator; direct-current to direct-current converter; Hardware-in-the-loop; Proportional integral differential; Renewable energy; single-ended primary-inductor converter; Voltage regulation

1 Introduction

Renewable energy sources are vital for ensuring a sustainable, clean, and secure energy future while reducing environmental impact. Power electronics enables high-efficiency conversion, robust control, and penetration of variable renewable energy resources into the power grid. The direct-current to direct-current (DC-DC) converters are used to convert one DC voltage level to another. Common types include boost, buck, Ćuk, buck-boost, and single-ended primary-inductor converters (SEPIC) [1–5]. A boost converter can provide only step-up, while a buck converter is helpful for step-down. Ćuk, Buck-boost, and SEPIC can do both. The basic converters, like buck and boost, have limitations. Buck produces pulsating input current, demanding larger input capacitors. Boost applies higher voltages to the load and possesses high output ripple. Buck-boost and Ćuk can invert output polarity, and hence, these converters will not be suitable for many practical applications.

The SEPIC converter offers buck-boost capability, non-inverting output, and low input current ripple. It supports both step-up and step-down operation while maintaining output polarity. It also reduces electromagnetic interference (EMI) and ripples, making it an ideal solution for renewable energy applications such as solar photovoltaic systems based on maximum power point tracking, charging an electric vehicle battery, despite requiring an extra capacitor and inductor. Unlike other converters, SEPIC provides voltage flexibility without inverting the output, and maintains

low current ripple on both input and output sides. These features make it especially useful in the applications of solar power extraction, but they require an objective function, which takes a long time to get accurate results [6–10]. Electric vehicles [11, 12], power factor correction [13], light-emitting diode (LED) driver [14], microgrid [15], renewable energy, and precision applications [16]. However, the SEPIC converters offer the following challenges:

(1) The circuit with two inductors and two capacitors exhibits nonlinear dynamic behavior.

(2) The non-minimum phase behavior of the duty cycle to output currents dynamics limits the achievable control bandwidth.

(3) The converter is highly sensitive to supply voltage and load variations, and parametric changes.

Designing a controller for a SEPIC converter is highly complex and must address these nonlinearities and dynamic variations. Therefore, the controller should have features like robustness to parameter variations, fast dynamic response, stability across operating conditions, and accurate tracking performance under disturbances to ensure smooth and reliable converter operation.

Linear techniques like linear quadratic regulator, PI, and proportional integral differential (PID) are applied for the control of the converters because of their simple design and easy implementation [17–20]. In these methods, the load voltage is compared with the reference in the outside loop, and the input current to the inductor is compared with the reference in the inner loop. However, these control techniques lack robustness to system uncertainties, and their long settling time and significant overshoot further limit their practical applications. Furthermore, the results are based on the ideal conditions rather than practical ones. To address the nonlinear behavior, various nonlinear and intelligent control techniques, such as neuro-synergetic control, have been used [21], fuzzy logic [22, 23], Lyapunov-based control [24], perturbation control [25], passivity-based control [26], etc, are applied to the SEPIC converter. Model predictive-based load voltage regulation is proposed in [27]. However, the primary concern with nonlinear controllers is their complex design and lack of robustness.

Sliding mode control (SMC) is commonly used as a powerful control technique with good performance and switching properties [28–32]. As a robust control scheme, SMC can resist external disturbance and parameter variation. Sliding mode control, a most suitable robust control for switching converters, is applied to the SEPIC converter [33–35]. Traditional SMC was developed for the SEPIC converter in continuous-time. However, in real-time, most of the controllers are implemented on digital processors or microcontrollers that work in discrete time (due to sampling and digital computation). Applying a continuous-time sliding mode controller (CTSMC) directly in a sampled system can lead to performance degradation or even instability. SMC is commonly used as a powerful control technique with good performance and switching properties [28–32]. As a robust control scheme, sliding mode control can resist external disturbance and parameter variation. Sliding mode control, a most suitable robust control for switching converters, is applied to the SEPIC converter [33–35].

Traditional SMC was developed for the SEPIC converter in continuous-time. However, in real-time, most of the controllers are implemented on digital processors or microcontrollers that work in discrete time (due to sampling and digital computation). Applying a CTSMC directly in a sampled system can lead to performance degradation or even instability. Table 1 presents a detailed comparison of different control strategies implemented for the SEPIC converter. The switching gain, chattering, and reaching law issues are essential factors for the optimum design of a discrete-time sliding mode controller (DTSMC).

Table 1. Comparison of various control methods of single-ended primary-inductor converter (SEPIC) converter

Ref.	Control	Disturbances	Nonlinearities	Parameter Variations	Discrete Time
[17]	Proportional integral differential (PID) control	✓	×	×	×
[24]	Lyapunov-based control	✓	✓	×	×
[34]	Sliding mode control	✓	✓	✓	×
[35]	Indirect sliding mode control	✓	✓	✓	×
	Proposed controller	✓	✓	✓	✓

In this paper, a modified discrete-time sliding mode controller (MDTSMC) strategy integrating a reference switching variable trajectory generator for controlling the SEPIC converter has been proposed. Aiming to mitigate the impact of nonlinearities and disturbances on the converter, a novel control strategy is applied to the SEPIC converter to guide its trajectories effectively [36]. The reference switching variable trajectory generator is produced externally and is immune to disturbances.

The main contributions of this paper can be summarized as follows:

(1) An MDTSMC law is developed for precise and robust SEPIC converter operation.

(2) The SEPIC converter proposed control law is simulated in MATLAB/Simulink and evaluated under multiple operating conditions. The simulation results confirm stable and precise performance over a broad range of operating

conditions.

(3) To further verify its practical feasibility, the proposed controller is evaluated in a hardware-in-the-loop (HIL) simulation under different operating conditions, and the results confirm its capability to maintain performance and robustness in real-time conditions.

2 Modeling of Sepsic Converter

The SEPIC converter mathematical model is derived using Kirchhoff's laws applied to its circuit topology. The inductors L_1 and L_2 include winding resistances r_{L_1} and r_{L_2} , and the capacitors are C_1 and C_2 . V_{in} is the input voltage and R_L is the load resistance. The obtained equations represent the energy transfer through the coupled inductors and capacitors, and they also show the influence of the switching control signals on the system states. The dynamic behavior of the SEPIC converter is characterized by differential equations presented in Eqs. (1)–(4):

$$\frac{di_{L_1}}{dt} = \frac{V_{in} - r_{L_1}i_{L_1}}{L_1} - (1-u) \left(\frac{V_{C_1} + V_{out}}{L_1} \right) \quad (1)$$

$$\frac{di_{L_2}}{dt} = \left(\frac{V_{C_1} - r_{L_2}i_{L_2}}{L_2} \right) u + (1-u) \left(\frac{-V_{out}}{L_2} \right) \quad (2)$$

$$\frac{dV_{C_1}}{dt} = \left(\frac{i_{L_1}}{C_1} \right) (1-u) - \left(\frac{i_{L_2}}{C_1} \right) u \quad (3)$$

$$\frac{dV_{out}}{dt} = \left(\frac{i_{L_1} + i_{L_2}}{C_2} \right) (1-u) - \frac{V_{out}}{C_2 R_L} \quad (4)$$

Here, the switching control variable is defined as:

$$u = \begin{cases} 1 & \text{S is ON} \\ 0 & \text{S is OFF} \end{cases}$$

So that u explicitly represents the duty cycle (switching state) in an averaged sense.

The continuous-time state-space representation of the SEPIC converter is given by:

$$\dot{x} = Ax + Bu + EdV_{in} \quad (5)$$

Let,

$$x = \begin{bmatrix} i_{L_1} \\ i_{L_2} \\ V_{C_1} \\ V_{out} \end{bmatrix}.$$

where, Ed represents external disturbances (e.g., winding resistances, parameter variations, unmodeled dynamics, load changes), and Ed distributes disturbances into the state equations. Now, by solving the above differential equations and state space equations, the resulting matrices can be given as follows:

$$A = \begin{bmatrix} -\frac{r_{L_1}}{L_1} & 0 & -\frac{(1-u)}{L_1} & -\frac{(1-u)}{L_1} \\ 0 & -\frac{r_{L_2}}{L_2} & \frac{u}{L_2} & -\frac{(1-u)}{L_2} \\ \frac{(1-u)}{C_1} & -\frac{u}{C_1} & 0 & 0 \\ \frac{(1-u)}{C_2} & \frac{(1-u)}{C_2} & 0 & -\frac{1}{C_2 R_L} \end{bmatrix}$$

$$B = \begin{bmatrix} \frac{(V_{C_1} + V_{C_2})}{L_1} \\ -\frac{(V_{C_1} + V_{C_2})}{L_2} \\ -\frac{(i_{L_1} + i_{L_2})}{C_1} \\ -\frac{(i_{L_1} + i_{L_2})}{C_2} \end{bmatrix}, Ed = \begin{bmatrix} \frac{1}{L_1} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

To facilitate implementation on digital processors or microcontrollers, the continuous-time model is now discretized using the Euler approximation approach with $T = 10 \mu s$. This results in discrete-time state-space

matrices. Thus, the converter dynamics given in Eq. (5) are now converted into the discrete time to become suitable for use with the reference switching variable trajectory generator. The equations are as follows:

$$x(k+1) = A_d x(k) + B_d u(k) + E_d d(k) V_{in} \quad (6)$$

where,

$$A_d = \begin{bmatrix} 1 - \frac{r_{L1}T}{L_1} & 0 & -\frac{(1-D)T}{L_1} & -\frac{(1-D)T}{L_1} \\ 0 & 1 - \frac{r_{L2}T}{L_2} & \frac{DT}{L_2} & -\frac{(1-D)T}{L_2} \\ \frac{(1-D)T}{C_1} & -\frac{DT}{C_1} & 1 & 0 \\ \frac{(1-D)T}{C_2} & \frac{(1-D)T}{C_2} & 0 & 1 - \frac{T}{C_2 R_L} \end{bmatrix}$$

$$B_d = \begin{bmatrix} \frac{(V_{C1}+V_{C2})}{L_1} \\ -\frac{(V_{C1}+V_{C2})}{L_2} \\ -\frac{(i_{L1}+i_{L2})}{C_1} \\ -\frac{(i_{L1}+i_{L2})}{C_2} \end{bmatrix} T, E_d = \begin{bmatrix} \frac{V_{in}}{L_1} \\ 0 \\ 0 \\ 0 \end{bmatrix} T$$

The duty cycle of the SEPIC converter in steady state conditions can be written as:

$$D = \frac{V_{out}^*}{V_{in} + V_{out}^*} \approx 0.6575 \quad (7)$$

The matrix B_d provides an approximate yet effective way to compute the discrete input matrix using the zero-order hold assumption. The high value of the sampling time would reduce system accuracy due to signal distortion, while the low value improves system accuracy but requires high computational resources. Since the SEPIC converter is a nonlinear system, this approach allows the discrete model to update B_d at each sampling instant. In practice, this discrete approximation supports accurate implementation of the control algorithm digitally without impacting the converter dynamics.

The matrices, A_d and B_d characterize the converter's sampled dynamics and provide the foundation for designing advanced discrete-time control strategies. By representing the SEPIC converter in this mathematical framework, precise regulation of its load voltage can be achieved in the event of input voltage variations and load disturbances.

3 Discrete-Time Sliding Mode Control

This paper presents a novel sliding-mode control approach designed for discrete-time systems subjected to disturbances, ensuring accurate tracking of a reference trajectory. The system can be written using the following equation:

$$x(k+1) = A_d x(k) + B_d u(k) + E_d d(k) \quad (8)$$

In SMC, the primary objective is to pull the states of the system to a predefined sliding surface (or hyperplane) and then keep them on that surface. This maintains robust performance even when disturbances and modeling uncertainties are present. The first stage of this process is known as the reaching stage, and during this stage, the system trajectories are guided from their initial positions toward the sliding surface. The control generated during this stage must guarantee that the trajectories reach the sliding surface in a finite time.

3.1 Sliding-Mode Objective and Sliding Hyperplane

In SMC, the objective is to force the system states to evolve on a predefined sliding surface, ensuring robustness against disturbances and uncertainties. The state tracking error is defined as:

$$e(k) = x_d(k) - x(k) \quad (9)$$

where, $x_d(k)$ is the desired or reference state.

The sliding hyperplane is introduced as:

$$\varphi(k) = C e(k) = C (x_d(k) - x(k)) = 0 \quad (10)$$

Here, C is chosen such that $(CB_d)^{-1} \neq 0$.

3.1.1 Disturbance modeling

Assume the disturbance enters the system as in Eq. (9). The effect of vector disturbance $d(k)$ on $\varphi(k)$ is $D(k)$. The influence of the disturbance on the sliding variable is defined as:

$$D(k) = C E_d d(k)$$

Let the mean disturbance effect be:

$$D_1 = \frac{1}{2} (CE_d d_{max} + CE_d d_{min}) \quad (11)$$

and the maximum deviation from this mean is:

$$D_2 = \frac{1}{2} (CE_d d_{max} - CE_d d_{min}) \quad (12)$$

Thus $|D(k) - D_1| \leq D_2$ for all admissible $d(k)$.

3.1.2 Reference switching variable trajectory generator

To improve robustness, a reference switching variable trajectory generator $\varphi_g(k)$ is generated externally using the nominal (undisturbed) model. This ensures that $\varphi_g(k)$ evolves toward zero according to a reference reaching law, independent of disturbances. The initial conditions are set to match:

$$\varphi_g(0) = \varphi(0) = C(x_d(0) - x(0)) \quad (13)$$

3.1.3 Reaching law

The proposed reaching law makes the actual sliding variable follow the generated reference trajectory while compensating for the average disturbance.

The reaching law for the system:

$$\varphi(k+1) = \varphi_g(k+1) - D(k) + D_1 - \sum_{i=0}^k \varphi(i) - \varphi_g(i) \quad (14)$$

The term $-D(k) + D_1$ compensates for the current disturbance deviation from its mean; it prevents accumulation of past disturbance effects that would otherwise widen the quasi-sliding band. The summation term $-\sum_{i=0}^k \varphi(i) - \varphi_g(i)$ is an integral-like correction used when disturbances are slowly varying.

By following this law, the system ensures convergence to the sliding surface, although satisfying constraints on control effort, rate of change, and chattering reduction. Essentially, the reaching law provides a structured rule for designing the control input so that the system states reliably enter the sliding regime, where the pros of SMC, including robustness and insensitivity to matched disturbances, can be fully exploited.

3.2 Control Law

The corresponding control law becomes:

$$u = (CB_d)^{-1} \left\{ Cx_d(k) - CA_d x(k) - \varphi_g(k+1) - D_1 + \sum_{i=0}^k \varphi(i) - \varphi_g(i) \right\} \quad (15)$$

The proposed MDTSMC law directs the system states toward the reference trajectory by applying a reaching law. This ensures that the sliding variable repeatedly crosses the sliding surface in a finite band.

3.2.1 Stability proof

This section shows that under mild parameter conditions, the controlled system enters a quasi-sliding mode in finite time and afterward $\varphi(k)$ stays in a bounded band.

Suppose the generator parameters satisfy:

$$\sigma_0 < D_2 \text{ and } \varepsilon > \frac{\sigma_0 D_2}{\sigma_0 - D_2} \quad (16)$$

Then the actual sliding variable $\varphi(k)$, governed by Eq. (13) (without the summation term), enters a quasi-sliding mode in finite time $k_0 \leq k_{0g} + 2$. For all $k \geq k_0$,

$$\text{sign}(\varphi(k)) = -\text{sign}(\varphi(k+1)), |\varphi(k)| \leq \varepsilon + D_2 \quad (17)$$

In the case of the slowly-varying disturbance assumption with the reaching law in Eq. (14),

$$|D(k+1) - D(k)| \leq \Delta \leq D_2$$

and the generator parameters are chosen such that

$$\sigma_0 > \Delta \text{ and } \varepsilon > \frac{\sigma_0 \Delta}{\sigma_0 - \Delta} \quad (18)$$

Then the quasi-sliding mode exists with the reduced band.

$$|\varphi(k)| \leq \varepsilon + \Delta \quad (19)$$

The proposed control law guarantees practical stability by maintaining the plant within a bounded quasi-sliding-mode band while ensuring convergence to the reference trajectory even when the disturbances and the uncertainties are present. Interested users can study the detailed proof of the stability by referring to Bartoszewicz and Adamiak [36].

The system stability using the proposed control law has been verified numerically, as shown in Figure 1. The sampling time is $10 \mu\text{s}$, sliding surface coefficient is 0.5, the disturbance bound is 0.02, and the convergence factor is 0.6. The result confirms that the system achieves boundedness within a quasi-sliding mode band (QSMB) under 50% load step disturbance (200Ω – 100Ω) at time 0.5 s with steady state 1.35 and 5.30, respectively.

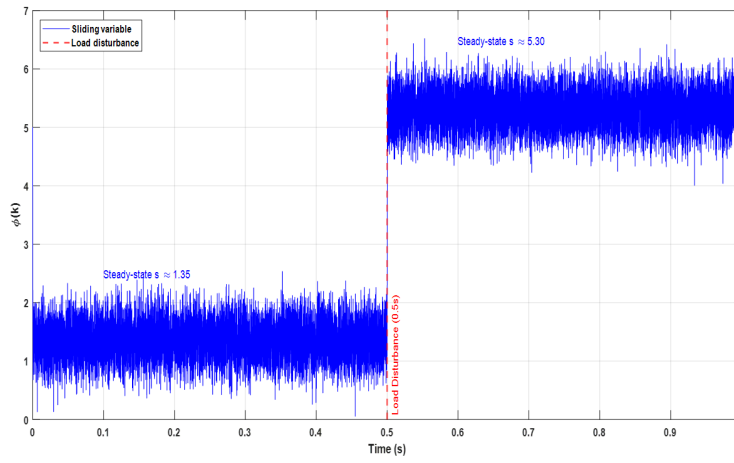


Figure 1. Sliding variable with 50% load disturbance at 0.5s

4 Discrete-Time Sliding Mode Control of Sepic Converter

The proposed system schematic is demonstrated in Figure 2. In the proposed DTSMC of a SEPIC converter, the voltage of the load is regulated by forcing the system states to track a reference trajectory through a discretely implemented sliding surface. The output voltage V_{out} is compared to the reference voltage V_r , and the resulting error is given to the PI controller, which generates a reference inductor current I_L^* . The difference of I_L^* and the inductor current (i_{L1}) provides the trajectory, which dynamically influences the control mechanism. This reference current, along with the measured capacitor voltages and inductor currents, defines the sliding variable, which characterizes the system tracking error in a multi-variable sense.

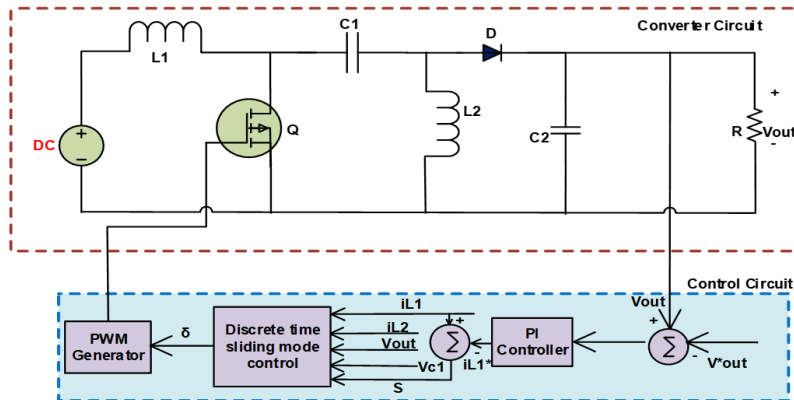


Figure 2. Proposed MDTSMC of the SEPIC converter

The MDTSMC algorithm then generates the required duty cycle by applying a reaching law that drives the sliding variable toward zero at each sampling instant. This guarantees sliding towards the surface.

When the sliding surface is achieved, the controller is able to hold the desired output voltage even when there are variations in the load, changes in input voltage and parameter uncertainties.

The full-discrete implementation of MDTSMC allows the smooth implementation of the discrete model, the dynamic properties of the converter, and the strong and well-performing voltage regulation even in conditions with fast variations of operation. The MDTSMC also varies the duty cycles to produce appropriate pulse width modulation (PWM) signals, which are effective in controlling the MOSFET and, therefore, energy transfer.

A workflow diagram of the proposed method for generating the duty cycle of the SEPIC converter using MDTSMC is shown in Figure 3. This diagram summarizes the step-by-step computational proposed approach for generating the duty cycle of the SEPIC converter using MDTSMC.

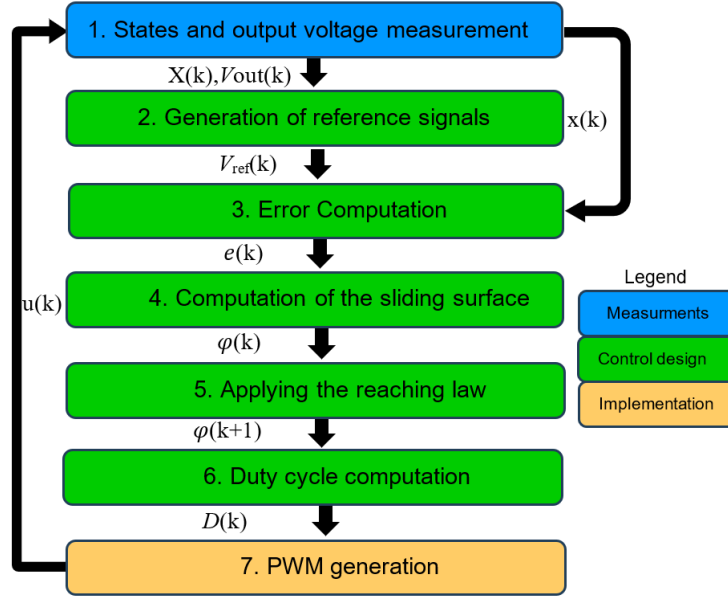


Figure 3. Process diagram for generating the duty cycle using the proposed method

Now, the following case study demonstrates the process of computing the duty cycles of the SEPIC converter using the proposed method. Note that to simplify the fourth-order structure of the SEPIC converter, only the voltage control loop will be considered.

Step 1: Define the converter parameters, which are summarized in Table 2.

Table 2. Details of the single-ended primary-inductor converter (SEPIC) converter and controller

Parameter	Value
Capacitors: C_1, C_2	330 μF
Input voltage: V_{in}	25 V
Resistances: r_{L1}, r_{L2}	0.14 Ω
Inductors: L_1, L_2	800 μH
Reference output voltage: V_{out}	48 V
Proportional Integral (PI) gain: k_p, k_i	2.5 and 1
Sampling time: T	10 μs
Switching frequency: f_s	100 kHz
Load 1, R_{L1}	200 Ω
Load 2, R_{L2}	100 Ω

Step 2: Measure the output voltage (voltage across C_2)

$$\text{Measure } x(k) = V_{out}(k) \quad (20)$$

Step 3: Find the values of the transition, input and disturbance matrices (A_d, B_d, E_d) in Eq. (6) with sampling time $T = 10\mu\text{s}$ and the results as follows:

$$A_d = \begin{bmatrix} 1 & 0 & -0.00428 & -0.00428 \\ 0 & 1 & 0.00821 & -0.00428 \\ 0.01038 & -0.0199 & 1 & 0 \\ 0.01038 & 0.01038 & 0 & 1 \end{bmatrix}$$

$$B_d = \begin{bmatrix} 0.9126 \\ 0.9126 \\ -0.0213 \\ -0.014 \end{bmatrix}$$

$$E_d = \begin{bmatrix} 0.3125 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

These matrices help to obtain the control law given in Eq. (15).

Step 4: Compute output voltage error and error difference

$$e(k) = V_{ref} - x(k) = 48 - 47.5 = 0.5 \quad (21)$$

Step 5: Design of sliding surface

This sliding surface will track the desired output voltage ($V_{out} = V_{C2}$)

$$\varphi(k) = c.e(k) = 0.45 * 0.5 = 0.225 \quad (22)$$

$$c = 0.45$$

Step 6: Compute reference switching variable trajectory generator

$$\varphi_g(k+1) = \lambda.\varphi(k) = 0.75 * 0.225 = 0.169 \quad (23)$$

With $\lambda = 0.75$ the SEPIC converter and its control system will softly approach the sliding surface, and therefore, the time of reaching the phase will be less.

Step 7: Design of the reaching law

To make the system track the generator perfectly under load change from 200Ω to 100Ω , the following

$$\varphi(k+1) = \varphi_g(k+1) - D(k) + D_1 - \sum_{i=0}^k \varphi(i) - \varphi_g(i)$$

Step 8: Compute the duty cycle $D(k)$

$$D(k) = \frac{\varphi(k+1) - c[A_d x(k) + E_{da}(k)V_{in} - V_{ref}]}{cB_d x(k)} = 0.6569$$

Step 9: Generate PWM

Step 10: Update previous values

Step 11: Waiting for the next sampling instant

5 Results and Discussion

The proposed MDTSMC approach for the SEPIC converter was first evaluated through detailed simulations in MATLAB/Simulink.

In the simulation environment, the controller's performance was analyzed under various operating scenarios, such as changes in reference voltage and load disturbances. This will ensure its robustness and dynamic response. Following the simulation studies, the proposed control strategy was experimentally verified in a HIL environment using the OPAL-RT platform. This HIL setup allowed realistic validation of the controller's real-time behavior, and this confirms its feasibility and effectiveness for practical implementation.

5.1 Simulation Results

The proposed MDTSMC method is validated by testing the converter under four different operating scenarios:

(1) A transition in reference voltage from buck mode to boost mode.

- (2) A transition in reference voltage from boost mode to buck mode.
- (3) A sudden variation in load resistance.
- (4) A sudden variation in the input voltage.

In the first case, the reference voltage is increased from an initial value of 25 V to 48 V in 0.1 s. This case study is beneficial for evaluating the controller's performance in boost mode. Figure 4 illustrates both the output voltage and the reference voltage waveforms. The results confirm that the proposed MDTSMC strategy effectively tracks the reference without noticeable overshoot or oscillations, proving its efficacy in achieving voltage regulation. Furthermore, the system achieves a settling time of less than 0.02 s, indicating a fast dynamic response.

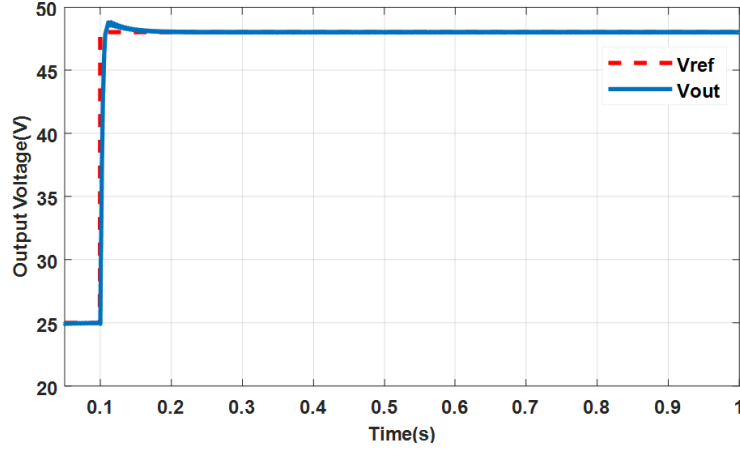


Figure 4. Output voltage in scenario 1

Figure 5 depicts the inductor current, showing a sudden increase at 0.1 s corresponding to the modification in the reference voltage. In the second testing scenario, the reference voltage is reduced from 40 V to 25 V at 0.1 s to verify the performance of the MDTSMC under buck conditions. This test assesses the system's ability to maintain stable operation and accurate voltage tracking when subjected to a significant downward change in reference.

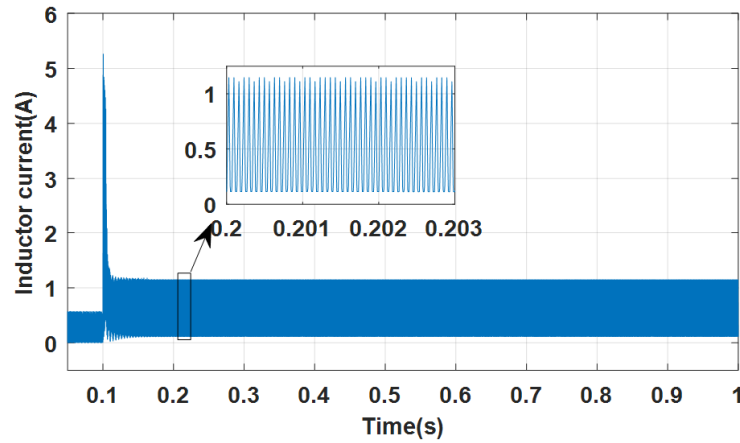


Figure 5. Inductor current in scenario 1

Figure 6 shows how the output voltage tracks the reference voltage in the case of sudden changes in the reference voltage. The load voltage closely follows the reference voltage despite the abrupt reduction, with no significant overshoot or oscillations.

Figure 7 demonstrates the inductor current under scenario 2. The current waveform clearly shows the current's adjustment to the reduced reference voltage, confirming the controller's capability to handle transient conditions effectively.

To verify the robustness of the proposed MDTSMC, a resistance load of 100 Ω is suddenly added to the existing load resistance of 200 Ω at 0.2 s. When the sudden load is applied, the converter quickly stabilizes the output back to 40 V, and this can be visualized in Figure 8. Even though the inductor current suddenly increases, the voltage variations are minimal, and this proves the robustness of the proposed MDTSMC method.

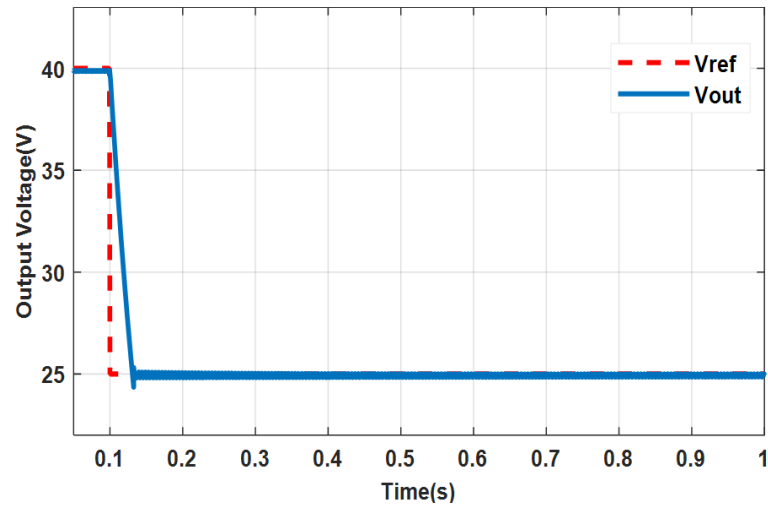


Figure 6. Output voltage in scenario 2

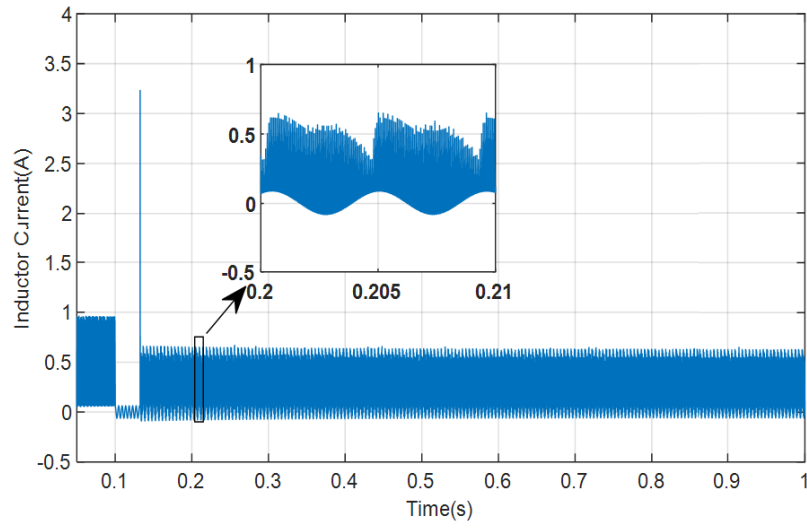


Figure 7. Inductor current in scenario 2

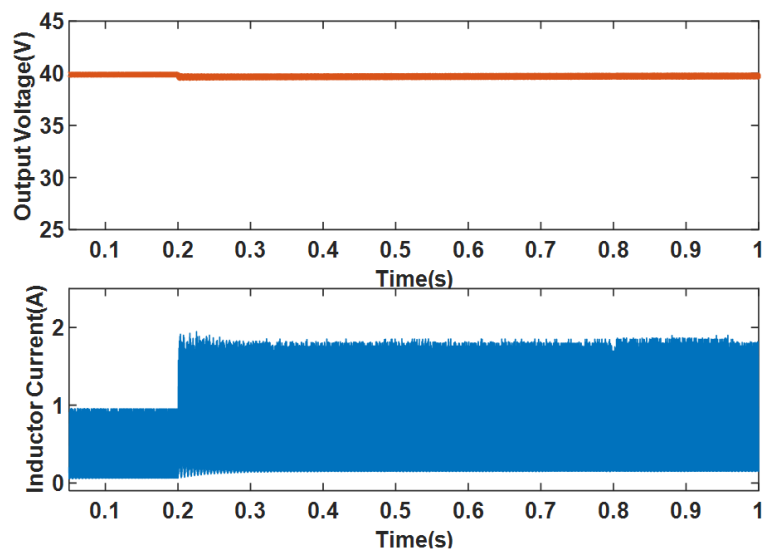


Figure 8. Output voltage and inductor current to a sudden change in load for scenario 3

To reflect a practical scenario in which the input voltage may change over time, the fourth testing scenario introduces a sudden variation in input voltage. Figure 9 illustrates both the output and the supply voltages. From the waveforms, it can be observed that even though the input voltage suddenly drops from 30 V to 25 V at 0.1 s, the impact on the output voltage is minimal, highlighting the robustness of the proposed sliding-mode controller.

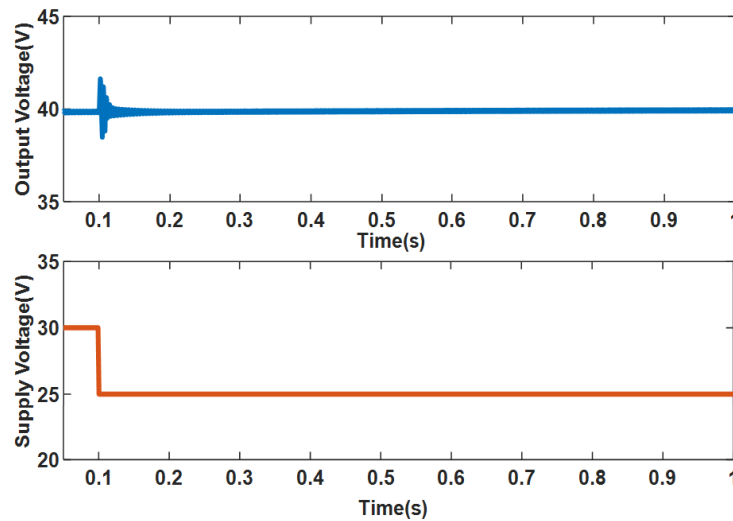


Figure 9. Output voltage and supply voltage waveforms for scenario 4

5.2 Comparison with Other Controls

The voltage response of the proposed SMC control approach is compared with a PID controller, a conventional SMC based on the signum function, and model predictive control (MPC) to investigate the performance, robustness, and effectiveness of each controller.

The dynamic behavior of these controllers with respect to the input and output reference voltage is shown in Figure 10. The gains of the PID controller are selected as: $K_p = 1$, $K_i = 0.5$, and $K_d = 0.01$. The gain of the SMC is set to 0.01. The PID controller shows the slowest response and significant steady-state error. In contrast, the basic SMC produces a faster dynamic response with high ripple due to the chattering effects, which will lead to an increase in the temperature of electrical components. The MPC performs better than PID and basic SMC but has higher overshoot, moderate robustness, and computational burden constraints compared with the proposed SMC.

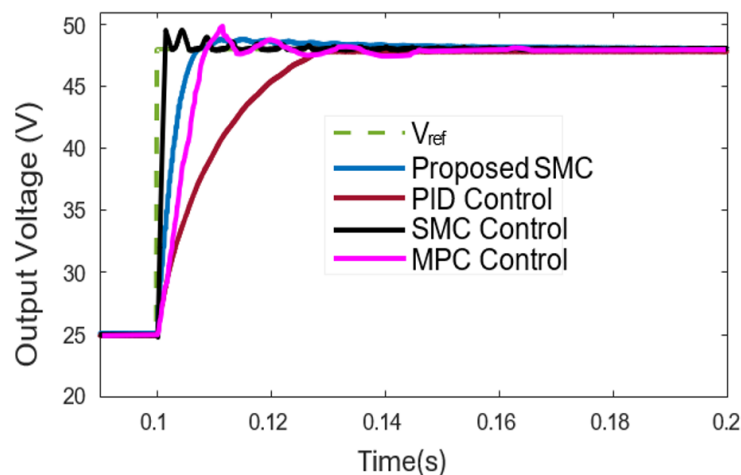


Figure 10. Comparison of the proposed SMC controller with PID control, conventional SMC, and MPC

Note: PID = proportional integral differential; SMC = sliding mode control; MPC = model predictive control

The dynamic performance of these controllers in terms of rise time, settling time, overshoot, and output ripple is summarized in Table 3. The results show that the overall performance of the proposed SMC is better than the others. However, PID has a lower computational burden than SMC and MPC. The SMC is a moderate in complexity with high disturbance rejection compared to PID and MPC.

Table 3. Performance comparison of PID, SMC, and MPC with the proposed method

Controller Type	Settling Time (t_s)	Overshoot (M_p)	Ripple (%)	Robustness
PID	21 ms	5.7%	1.8%	Moderate
Conventional SMC	8 ms	2.6%	5.3%	High
MPC	10 ms	2.1%	1.2%	Low
Proposed SMC	7 ms	1.9%	4.6%	High

Note: PID = proportional integral differential; SMC = sliding mode control; MPC = model predictive control.

5.3 Hardware in Loop Results

The HIL simulation results were thoroughly evaluated using the OPAL-RT platform to validate the effectiveness of the boost mode operation of the converter. The Y-axis scale corresponds to 10 volts per division, while the X-axis represents 0.1 s per division. The simulation shown in Figure 11 demonstrated that the converter successfully increased the input voltage from 48 V to 66 V, achieving the desired voltage boost with satisfactory performance.

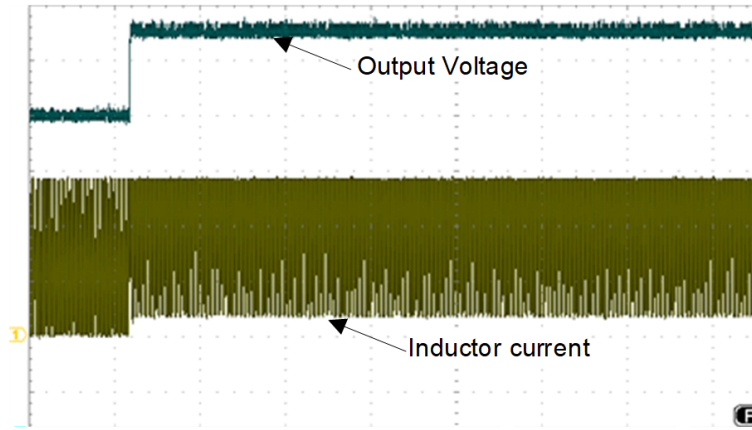


Figure 11. Dynamic response of the output voltage and the inductor current to a reference voltage step change in boost mode using hardware-in-the-loop (HIL)

Upon applying the load at approximately 0.6 s, a brief disturbance was observed in the output voltage waveform, likely due to the sudden increase in current demand. However, the controller adjusts the duty cycle to this disturbance, and hence the output voltage quickly settles back to the target level of 66 V as illustrated in Figure 12. It demonstrates the effective dynamic regulation and robust control behavior under load variations.

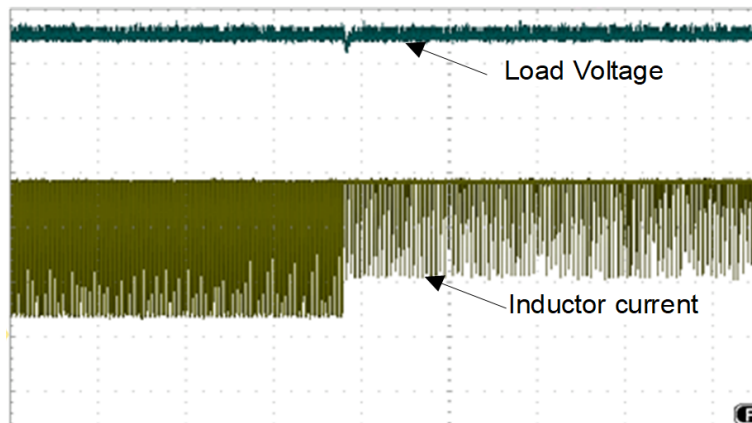


Figure 12. Output load voltage and inductor current response to a rapid variation in load resistance using hardware-in-the-loop (HIL)

Now, to evaluate the performance of the MDTSMC under buck, the reference voltage is suddenly increased to 16 V, and the results of the HIL are illustrated in Figure 13. The results also demonstrate the efficacy of the controller with a fast-settling time.

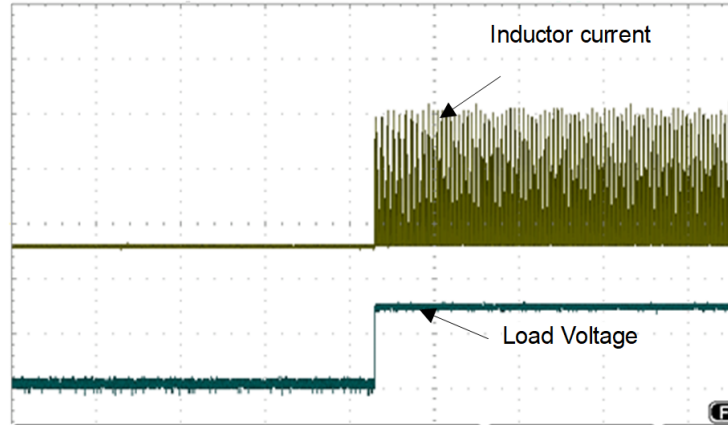


Figure 13. Response of the converter under buck mode using hardware-in-the-loop (HIL)

In the final test scenario, a load resistance is suddenly applied while the SEPIC converter operates in buck mode. Figure 14 depicts the response of the converter in this scenario. Even though the load is suddenly increased, the impact of the disturbance on the functioning of the converter is negligible.

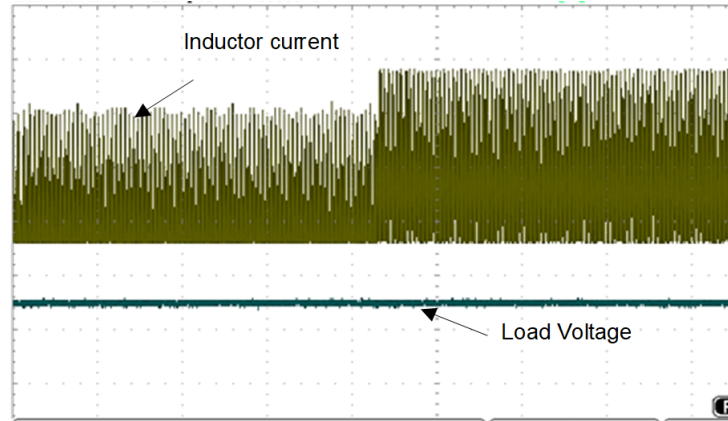


Figure 14. Response of the converter to a sudden increase in the load at buck mode using hardware-in-the-loop (HIL)

Table 4 summarizes the performance of the experimental result using HIL with respect to steady state error and overshoot. The steady-state error falls roughly between 3.10% and 3.92%, while overshoot ranges from about 4.50% to 6.25%. These numbers indicate a well-damped, reasonably fast response: overshoot is modest, and the steady-state accuracy approaches zero.

Table 4. Summary of the hardware-in-the-loop (HIL) performance

	Steady State Error (%)	Overshoot (%)
Figure 9	3.23	5.13
Figure 10	3.65	6.25
Figure 11	3.92	4.50
Figure 12	3.10	5.87

6 Conclusions

In this paper, an MDTSMC has been utilized for performance improvement of a SEPIC converter with minimum disturbance effects and to reduce the quasi-sliding-mode bandwidth. The proposed controller demonstrated excellent performance in MATLAB simulations across four different operating scenarios, effectively handling input and load variations. To validate its practical applicability, HIL simulations using OPAL-RT were conducted, confirming the controller's robust voltage regulation, fast dynamic response, and strong disturbance rejection. However, the

chattering, computational problems, sampling and parameter sensitivity that affect reachability to a nonzero steady-state are the main constraints of the SMC discretization approach. The proposed DTSMC strategy provides an effective solution for real-time digital control of power converters, delivering high performance and reliability across a wide range of practical conditions. The suggested MDTSMC plan is a valuable framework that can be used to introduce real-time, digital control of power converters and achieve high performance and reliability in a broad selection of experimentally applicable conditions. The suggested MDTSMC may be scaled to multi-phase SEPIC topologies and enhanced with more sophisticated methods of observer-based performance in more complicated power electronic systems. Moreover, it is the best option for renewable energy applications.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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Nomenclature

A, B, E	Continuous time state transition, input and disturbance matrices
A_d, B_d, E_d	Discrete time state transition, input and disturbance matrices
C_1, C_2	Capacitance
c	Parameter of the sliding surface
CTSMC	Continuous-time sliding mode controller
D	Duty cycle
D_1	Average value of $D(k)$
D_2	Maximum value of $D(k)$
$d_{\max}$	Maximum disturbance
$d_{\min}$	Minimum disturbance
MDTSMC	Modified discrete-time sliding mode control
$d(t)$	External disturbance
E	State tracking error
$e(k)$	Error at time (k)
$e(k+1)$	Error at time (k+1)
f_s	Switching frequency
HIL	Hardware-in-the-loop
I_L^*	Reference inductor current
i_{L1}, i_{L2}	Current through inductor L_1, L_2
K	Sampling instant
K_p	Proportional gain
K_i	Integral gain
L_1, L_2	Inductance
MPC	Model predictive control
PI	Proportional Integral
PWM	Pulse width modulation
q	Gain of reachin law
R_L	Load resistance
r_{L1}, r_{L2}	Winding resistances
R_{L1}, R_{L2}	Resistance loads 1 and 2
S	Switch
SEPIC	Single-Ended Primary-Inductor Converter
SMC	Sliding mode control
T	Sampling period
u	Switching control variable
V_{C1}, V_{C2}	Voltage across capacitor C_1, C_2
V_{in}	Input voltage
V_{out}	Output voltage
V_r	Reference voltage
X	State vector
x_d	Desired state
$\varphi_g(k)$	Reference switching variable trajectory generator
$\varphi(k)$	Actual sliding variable
λ	Transition gain
ε	Reaching speed
σ_0	constant